

Mohamed Thouseef

TEST AUTOMATION ENGINEER — EMBEDDED SOFTWARE & HARDWARE VALIDATION

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linkedin.com/in/mohamedthouseef07 | **Nationality:** Indian | Open to relocation

Professional Summary

Test Automation Engineer with 4.5 years of experience in embedded software and hardware validation for automotive infotainment systems. Experienced in designing scalable automation frameworks using Python, C, Robot Framework, Appium, and Jenkins for CI/CD environments. Strong expertise in CAN, CAN FD, UDS, SPI, I2C, UART, Hardware-in-the-Loop (HIL) testing, root cause analysis, and embedded diagnostics using CANoe, oscilloscopes, and logic analyzers. Proven record of reducing validation time, improving automation coverage, and delivering high-quality software releases through cross-functional collaboration

Technical Expertise

PROGRAMMING: C | Embedded C | Python (advanced) | CAPL

AUTOMATION FRAMEWORKS & TOOLS: Robot Framework | Appium | Automation Framework Development | Automated Test Case Development | BDD | Cucumber | NI Vision | Tesseract OCR | REST API | API Testing

EMBEDDED SYSTEMS & FIRMWARE VALIDATION: STM32 (ARM Cortex-M) | Device Driver Development | Firmware Validation & Debugging | Embedded Software Validation | GPIO | Interrupts | Timers | ADC | RTOS Concepts

HARDWARE DEBUGGING & TEST INSTRUMENTS: CANoe | Logic Analyzer | Oscilloscope | JTAG | Trace Analysis | Tera Term

COMMUNICATION PROTOCOLS: CAN | CAN FD | UDS (ISO 14229) | SPI | I2C | UART

TESTING METHODOLOGIES: System-Level Testing | Functional Testing | Regression Testing | Stability & Stress Testing | Hardware-in-the-Loop (HIL) | Protocol Compliance Testing |

DEVOPS & LIFECYCLE: Jenkins | GiLab | CI/CD | Git | Code Review & Merge Request Review

PROJECT MANAGEMENT: JIRA | Agile/Scrum | Test Planning & Documentation | SDLC | Functional Test Strategy | Test Case Design | Automated Test Equipment (ATE) | Program Release Process

Professional Experience

System Validation Engineer – II (Test Program & Automation) **Visteon Corporation**

April 2022 – Present | Chennai, India

- Owned end-to-end **functional test program development** for automotive infotainment ECUs — defined test strategies, authored test cases, and managed program release process across OEMs.
- Reduced automated test execution time by **50% across 500+ daily CI/CD builds** by designing Python-based test automation frameworks integrated with Jenkins and GitLab.
- Improved CAN/UDS diagnostic fault detection by 25% by developing reusable **Python and C automation** libraries for **protocol validation, root cause analysis, and regression testing**.
- Automated Sleep Power Mode and Dark Current measurement validation** on the bench, catching 50+ standby power-leakage defects before production sign-off and cutting reliance on manual validation effort by 30%.
- Increased defect detection accuracy by 50% by implementing an **OCR-based computer vision automation** solution using **NI Vision and Tesseract** for display validation.
- Built a bench-level **Crank Profile Automation Suite** using **programmable power supplies and oscilloscope**-verified voltage profiling to replicate engine-crank voltage transients without a physical vehicle
- Generated 450+ regression test cases** and improved recurring defect coverage by 60% by implementing an **AI-assisted test case generation** workflow using historical defect data.
- Delivered on-time validation** for multiple automotive infotainment programs by **collaborating with software, hardware, and system engineering** teams to define test strategy, ensure requirement traceability, and manage the defect lifecycle

Awards & Recognition

- Visteon Silver Award (Aug 2026) – Outstanding Accomplishment & Business Impact
- Visteon Silver Award (Jan 2026) – Outstanding Accomplishment & Business Impact
- Visteon Silver Award (March 2024) – Dedication & Exceptional Efforts Towards Business Success

Key Projects & Achievements

Mahindra XUV700 - System Reliability & Power Mode Validation

Technologies Used: ROBOT Framework, Python, NI – Vision Acquisition, UDS, JIRA, Jenkins, GIT

- Developed and maintained automated system-level test suites covering power mode transitions, boot stability, and core infotainment functionality, improving validation efficiency and consistency.
- Designed and implemented an **automated power mode validation solution** using **NI Vision and Tesseract OCR**, identifying **1,200+ display anomalies** with **50% higher detection accuracy** and preventing **15 potential field issues** before vehicle launch
- Built virtual ECU simulation frameworks enabling CAN Network Validation for External ECUs – Climate, Drive Modes and BCM Modules

TATA Punch - USB-Based Automation & Infotainment Mobile Integration

Technologies Used: Appium, Jenkins, Git, CI/CD Pipelines

- Developed an automated software flashing pipeline that reduced flashing time by 40% and eliminated 80% of manual intervention, significantly improving testing throughput in high-volume validation environments
- Automated validation of Bluetooth, Android Auto, and Apple CarPlay functionalities across 10+ mobile device variants, achieving 95% test coverage while reducing manual testing effort by 30%

Skoda Kylaq - Protocol Compliance & Diagnostics Automation

Technologies Used: CAPL, C, ProveTech, Vector CAN, UDS, Canoe Simulations

- Developed and executed 200+ automated UDS diagnostic test cases covering services 0x10, 0x22, 0x2E, 0x2F, 0x19 and 0x31 using C and ProveTech, reducing manual validation effort by 40%
- Performed End-of-Line (EOL) diagnostic validation to ensure ISO 14229 compliance, resulting in a 25% improvement in fault detection accuracy and enhanced diagnostic reliability.

SPI EEPROM and I2C RTC Driver Implementation (STM32) – Self-initiated projects

Technologies Used: STM32CubeIDE, STM32CubeMX, I2C/SPI/UART

- Implemented SPI-based EEPROM driver with software-controlled chip select and single-byte read/write operations.
- Integrated I2C-based DS1307 RTC with register-level read/write and BCD time conversion.
- Designed interrupt driven UART logging for system monitoring and debugging.
- Validated timing, bus stability, and error handling using logic analyzer and UART logs.

EDUCATION & CERTIFICATIONS

B.E. Electronics & Instrumentation Engineering - 2017 – 2021

Jeppiaar Engineering College, Anna University - 74.5%

Relevant coursework: semiconductor device physics, electronic circuits, signal conditioning, microcontroller systems, instrumentation & measurement.

Advanced Embedded Systems - Sep 2021 – Mar 2022

Vector India: C, Embedded C, I2C/SPI, Linux, Microcontrollers

Real-Time Embedded Systems - Jan 2026 – Present

Coursera: RTOS, VxWorks, POSIX threads, CPU scheduling